Registration no:

Total Number of Pages: 02

B.TECH
PCEE4402

8th Semester Regular / Back Examination 2015-16
POWER SYSTEM PROTECTION
BRANCH(S): EEE, ELECTRICAL
Time: 3 Hours
Max Marks: 70
Q.CODE:W132

Answer Question No.1 which is compulsory and any five from the rest.
The figures in the right hand margin indicate marks.

Q1 Answer the following questions: **(2 x 10)**

- **a)** What is surge absorber? Explain How surge absorber differ from surge diverter?
- **b)** What type relay is best suited for long distance very high voltage transmission lines?
- **c)** State the advantages and disadvantage of electromagnetic relays.
- **d)** Give advantages of neutral grounding.
- **e)** Derive expressions for restriking voltage and RRRV
- **f)** What are the various faults that would affect an alternator?
- **g)** Give the limitations of Merz Price protection.
- **h)** What are the advantages of static relay over electromagnetic relay?
- **i)** Write any two properties of contact material used in vacuum circuit breaker?
- **j)** What is isolator and explain its requirements.

Q2
- **a)** Differentiate between primary and backup protection. **(5)**
- **b)** List various building blocks that constitute the static relay and give brief description about each of the blocks. **(5)**

Q3
- **a)** What are the physical chemical and dielectric properties of SF_6 Gas. **(5)**
- **b)** What is the principle of distance relaying? Explain with neat diagram. Definite distance relay. **(5)**

Q4 Describe the construction and principle of operation of induction type directional over-current relay. Explain IDMT characteristics and how they are obtained in an induction type relay. **(10)**

Q5
- **a)** Explain with neat circuit diagram the fault bus protection scheme. **(5)**
- **b)** A 3phase transformer of 220/11000 line volts is connected in Y/Δ. The protective transformers on 220 Volts side have a current ratio of 600/5 Amps. What should be the CT ratio on 11000 Volts side?. **(5)**

Q6 a) State the sequence of operation of circuit breaker, isolator and earthing switch, while closing. **(5)**

b) Briefly explain different methods of testing of circuit breakers? **(5)**

Q7 With neat sketch and phasor diagram explain how a negative phase sequence relay is employed for protection of electrical power system? **(10)**

Q8 Write notes on any two. **(5 x 2)**

a) MHO relay

b) DC circuit breaker

c) Buchholz's relay

d) Resistance switching.

Registration No :

Total Number of Pages : 02

B.Tech.
PCEE4402

8th Semester Regular / Back Examination 2017-18
POWER SYSTEM PROTECTION
BRANCH : EEE, ELECTRICAL
Time : 3 Hours
Max Marks : 70
Q.CODE : C290
Answer Question No.1 which is compulsory and any five from the rest.
The figures in the right hand margin indicate marks.

Q1. **Answer the following questions :** **(2 x 10)**

- **a)** What is the meaning of under reach of a relay?
- **b)** What is arc resistance? How is it found out?
- **c)** If a certain relay has a plug setting of 1.5 A, what will be the value of PSM for a fault current of 1000 A, provided the CT has a ratio of 400/1?
- **d)** What is an incipient fault? What is its adverse effect?
- **e)** What do you understand by time and distance grading? Give an example.
- **f)** What is the difference between circulating current principle type and balanced voltage type relays?
- **g)** What is the role of summation transformer?
- **h)** A particular fault has the sequence currents I_1 = -1.5 p.u., I_2 = 0.8 p.u., and I_0 = 0.7 p.u. What type of fault has occurred in the system?
- **i)** What do you understand by current chopping?
- **j)** What do you understand by RRRV? Give an expression.

Q2. **a)** How to detect an inter-turn fault in case of generator? Draw circuit diagrams to illustrate it. **(5)**

b) A three phase transformer rated 33/11 kV, 500 kVA Δ/Y, has a CT rating of 20/1 on the H.V. side in each phase. What will be the rating of the CT on the L.V. side so that it won't respond to any through fault? **(5)**

Q3. **a)** Illustrate the working of a numerical relay with proper block diagram. **(5)**

b) What do you understand by cosine type phase comparator? Explain its working. **(5)**

Q4. **a)** What do you understand by zones of protection in reference to distance protection? How would you implement the mho relay in context to different zones of protection? **(5)**

b) Find the expression for finding out the region of trip operation for offset mho relay **(5)**

Q5. **a)** How would you implement the static relays for different distance protection schemes? **(5)**

b) What is earth fault protection? What are the different methods of implementation of the same? **(5)**

Q6. **a)** What protection schemes are implemented for prime mover failure and loss of excitation in case of generators? **(5)**

b) What is auto reclosing? Explain briefly. **(5)**

Q7. Find the expression of fault current in a) L-L-G and b) L-G for a system with neutral impedance and fault impedance present. **(10)**

Q8. **Write short answer on any TWO :** **(5 x 2)**

a) Inrush current protection in transformer

b) Vacuum Circuit breaker

c) Translay scheme of feeder protection

d) Induction type relays and its torque equation

Registration No : | | | | | | | | | | |

Total Number of Pages : 02 **B.Tech**

PCEE4402

8^{th} Semester Back Examination 2018-19

POWER SYSTEM PROTECTION

BRANCH : EEE, ELECTRICAL

Time : 3 Hours

Max Marks : 70

Q.CODE : F080

Answer Question No.1 which is compulsory and any FIVE from the rest.

The figures in the right-hand margin indicate marks.

Q1 **Answer the following questions :** **(2 x 10)**

- **a)** Define the terms : i. burden ii. Operating time of a relay.
- **b)** Define bias in a biased differential relay.
- **c)** If the zero-sequence current of a generator for line to ground fault is j2.4 p.u., then what is the current through the neutral during the fault? Answer, with reasoning.
- **d)** Draw the zero-sequence network for a Δ-Δ connected 3-phase transformer.
- **e)** Draw the sequence impedance diagram for a single line to ground fault near a generator with no load.
- **f)** State the reasons for using IDMT relays for overcurrent protection.
- **g)** Draw the offset mho relay characteristics.
- **h)** Briefly explain, what is RRRV?
- **i)** Keeping in view the cost and overall effectiveness, which of the the following CBs is best suited for capacitor bank switching and why? CBs are vaccum, airblast, SF_6 and oil type.
- **j)** What is the role of anti-aliasing filter in a digital relay?

Q2 **a)** Derive an expression for total complex power in a 3-phase system in terms of symmetrical components of voltage and currents. **(5)**

b) Derive an expression for fault current line-to-line fault by symmetrical components method. **(5)**

Q3 **a)** Explain the working and constructional features of a reactance relay. **(5)**

b) Explain the duality between an amplitude and a phase comparator. **(5)**

Q4 **a)** Enumerate the various types of protection used in an alternator. Describe one method for the protection of stator of an alternator against faults. **(5)**

b) Discuss the general principle of operation of a transformer differential protection scheme? And explain percentage differential relay? **(5)**

Q5 **a)** For a 132-kV system, the reactance and capacitance up to the location of the circuit breaker is 3 ohms and 0.015μF, respectively. Calculate the following: **(5)**

i. The frequency of transient oscillation.

ii. The maximum value of restriking voltage across the contacts of the circuit breaker.

iii. The maximum value of RRRV.

b) What is resistance switching? Find an expression for critical resistance. **(5)**

Q6 Explain with necessary diagram the principle of operation of a distance protection scheme. How a 3 zone stepped units help to achieve fast discrimination? **(10)**

Q7 Draw the block diagram of a numerical relay and explain? Also, critically compare it with the electromagnetic relays. **(10)**

Q8 **Write short answer on any TWO :** **(5 x 2)**

a) Transformer Protection

b) SF_6 circuit breaker

c) Air Blast circuit breaker

Registration No :

Total Number of Pages : 02 **B.Tech**
PCEE4402

4^{th} Year Special Examination 2018-19
POWER SYSTEM PROTECTION
BRANCH : EEE, ELECTRICAL
Time: 3 Hours
Max Marks : 70
Q.CODE : G008
Answer Question No.1 which is compulsory and any FIVE from the rest.
The figures in the right-hand margin indicate marks.

Q1 **Answer the following questions :** **(2 x 10)**

a) Define the terms:
  a. burden
  b. operating time of a relay.

b) Write an expression for total complex power in a 3-phase system in terms of symmetrical components of voltage and currents.

c) Define bias in a biased differential relay.

d) The positive sequence component of voltage at the point of fault is zero. Identify the type of fault.

e) Draw the zero-sequence network for a Y-Δ connected 3-phase transformer.

f) Draw the sequence impedance diagram for a single line to ground fault near a generator with no load.

g) Why IDMT relays are used for over current protection?

h) Draw the characteristics of an impedance relay.

i) Briefly explain, what is RRRV?

j) Briefly state the importance of anti-aliasing filter in a digital relay.

Q2 **a)** The three sequence components of bus voltage for a given bus are: V_0=-0.105pu, V_1=0.953pu and V_2=-0.230 pu. Determine the three phase voltages. **(5)**

b) Develop an equivalent network showing the interconnection of sequence networks to simulate a double line to ground fault. **(5)**

Q3 **a)** Write the synthesis of Mho relay using an Amplitude Comparator. **(5)**

b) Explain the duality between an amplitude and a phase comparator. **(5)**

Q4 **a)** Enumerate the various types of protection used in an alternator. Describe one method for the protection of stator of an alternator against faults. **(5)**

b) Discuss the general principle of operation of a transformer differential protection scheme? And explain percentage differential relay? **(5)**

Q5 **a)** For a 132-kV system, the reactance and capacitance up to the location of the circuit breaker is 3 ohms and 0.015µF, respectively. Calculate the following : **(5)**
  a. The frequency of transient oscillation.
  b. The maximum value of restriking voltage across the contacts of the circuit breaker.
  c. The maximum value of RRRV.

b) What is resistance switching? Find an expression for critical resistance. **(5)**

Q6 Explain with necessary diagram the principle of operation of a distance protection scheme. How a 3 zone stepped units help to achieve fast discrimination? **(10)**

Q7

a) Draw the block diagram of a numerical relay and explain its working principle. **(5)**

b) Discuss about numerical over current relay with the help of a flow chart. **(5)**

Q8 **Write short answer on any TWO :** **(5 x 2)**

a) Static Relays

b) SF_6 circuit breaker

c) Air Break circuit breaker

Registration No : | | | | | | | | | | |

Total Number of Pages : 02

BTech
REL6D001

6th Semester Regular / Back Examination: 2021-22
ELECTRIC POWER SYSTEM PROTECTION
BRANCH(S): ELECTRICAL
Time : 3 Hour
Max Marks : 100
Q.Code : J317
Answer Question No.1 (Part-1) which is compulsory, any eight from Part-II and any two from Part-III.
The figures in the right hand margin indicate marks.

Part-I

Q1 **Answer the following questions :** **(2 x 10)**

a) What do you mean by under reach and over reach of a relay?
b) What is the significance of sensitivity in the protection systems?
c) What do you mean by TSM? How it affects the performance of a relay?
d) What is the impact of fault clearing time on the operation of protective devices?
e) What is meant by current chopping?
f) What is RRRV? Explain its significance..
g) Draw the zero-sequence network for a Δ- Y connected 3-phase transformer.
h) Identify the type of fault if the positive sequence component of voltage at the point of fault is zero.
i) What is the effect of CT saturation on the design of protective relaying systems?
j) What is meant by auto-reclosing action in power system protection?

Part-II

Q2 **Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve)** **(6 × 8)**

a) Derive the general equations of magnitude comparator incorporating the block diagram and phasor diagram.
b) Discuss the working principle of coincidence type phase comparator? Derive the mathematical expressions related to it.
c) The current rating of an Overcurrent relay is 5A. The relay has a plug setting of 150% and the time setting (TMS) of 0.4. The CT ratio is 400/5. Determine the operating time of the relay for a fault current of 6000A. At TMS=1, operating time at various PSM are given in the below table.

PSM	2	4	5	8	10	20
Operating Time (Sec)	10	5	4	3	2.8	2.4

d) Draw the schematic diagram of a numerical relay and explain the functions of various components.

e) With a neat sketch, explain the construction and working principle of a reverse power or directional relay.

f) What is an impedance relay? Explain the operating principle, torque equation and operating characteristics for impedance relay.

g) Define the term 'pilot' with reference to power line protection. List the different types of wire pilot protection schemes and explain for any one of the scheme

h) With a neat sketch explain the working of differential protection of 3-Phase circuits and balanced (opposed) voltage differential protection

i) What are the protective devices employed for the protection of an alternator against i) Overvoltage. ii) Overspeed ? Discuss them in brief.

j) Write a short note on "Buchholz relay" used for the protection of transformers to detect incipient fault.

k) With a neat sketch, explain the recovery rate theory and energy balance theory of arc interruption in a circuit breaker.

l) With a neat sketch and waveform explain the interruption of capacitive current.

Part-III

Only Long Answer Type Questions (Answer Any Two out of Four)

Q3 Describe the construction and working principle of an Induction Cup Type Electromagnetic Relay. Derive the torque equation and discuss its advantages and disadvantages. **(16)**

Q4 Explain with necessary diagram the principle of operation of a distance protection scheme. How a 3 zone stepped units help to achieve fast discrimination? **(16)**

Q5 Discuss the construction and general principle of operation of a transformer differential protection scheme? Explain percentage biasing of differential relay? **(16)**

Q6 Explain the construction and working principle of SF6 circuit breaker. Discuss the advantages and disadvantages of SF6 circuit breaker. **(16)**

Registration No :

Total Number of Pages : 03

Course: B.Tech
Sub_Code: REL6D001

6th Semester Regular/Back Examination: 2022-23
SUBJECT: Electric Power System Protection
BRANCH(S): ELECTRICAL
Time : 3 Hour
Max Marks : 100
Q.Code : M307
Answer Question No.1 (Part-1) which is compulsory, any eight from Part-II and any two from Part-III.
The figures in the right hand margin indicate marks.

Part-I

Q1 Answer the following questions: **(2 x 10)**

a) What is the difference between a fuse and a relay?
b) Why are differential relays more sensitive than over current relays?
c) What do you mean by pick-up value of a relay?
d) What is Plug-setting multiplier and Time-setting multiplier?
e) What should be the essential qualities of an ideal relay?
f) What is the difference between electromagnetic and static relays?
g) What is the need for short circuit studies or fault analysis?
h) What are the advantages and disadvantages of an Air blast circuit breaker?
i) Explain what you mean by RRRV. Write its mathematical expression.
j) What is meant by resistance switching?

Part-II

Q2 Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve) **(6 × 8)**

a) Explain the roles of primary and back-up relays in designing an effective protection strategy in power system.
b) Discuss the various components used in protective system and explain the basic principle of operation of electric power system protection.
c) Write a short note on the sequence component and sequence impedance.
d) The positive negative and zero sequence impedance of 125 MVA, Three phase, 15.5 KV Star grounded, 50 Hz generator are j0.1 pu, j0.05 pu, j0.01 pu respectively on machine rating base. The machine is unloaded and working at rated terminal voltage. If grounding impedance of generator is j0.01 pu, then calculate magnitude of fault current in KA, for a B-phase to ground.
e) With neat diagram, explain the principle of zones of protection in power system. Explain the application of zones protection schemes with comments on its merits and demerits.
f) A 3-phase, 20 MVA, 11 kV star connected alternator is protected by Merz-Price circulating current system. The star point is earthed through a resistance of 5 ohms. If the CTs have a ratio of 1000/5 and the relay is set to operate when there is an out of balance current of 1·5 A, calculate : (i) the percentage of each phase of the stator winding which is unprotected. (ii) the minimum value of earthling resistance to protect 90% of the winding.
g) Explain the principle of operation of directional over current relays with mathematical background. Explain what is meant by directional properties of the relays.

h) Describe the operating principles of impedance type of relays with explanation of the impedance characteristics.

i) Explain the pilot protection scheme used in power system. Describe the different type of pilots with applications.

j) With a neat diagram explain the construction and principle of operation of Buchholz relays. Comment on the challenges on its design and operation.

k) Write a short note on Coincidence type phase comparator.

l) Write a short note on the vacuum type circuit breakers.

Part-III
Only Long Answer Type Questions (Answer Any Two out of Four)

Q3 For the single line diagram of a 3-bus system as shown in Fig. 1, the sequence data for transmission lines and generators are given in Table 1. If a single line to ground fault occurs at F, calculate the fault current. When the fault impedance is j0.1, calculate the value of the fault current? **(16)**

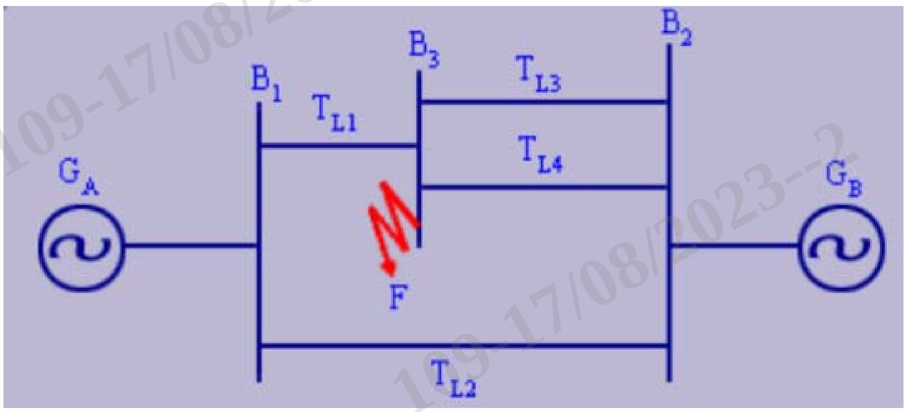

Fig. 1

Description	Sequence Data in pu		
	Zero	Positive	Negative
Generator - A	j0.03	j0.25	j0.15
Generator - B	j0.02	j0.20	j0.12
Transmission Line 1	j0.14	j0.08	j0.08
Transmission Line 2	j0.17	j0.13	j0.13
Transmission Line 3	j0.10	j0.06	j0.06
Transmission Line 4	j0.12	j0.06	j0.06

Table 1

Q4 Explain the principle of operation of induction type relays and derive the mathematical equations for representing these type of relays. Discuss the opportunities and challenges faced for this type relays. **(16)**

Q5 Elaborate with neat diagram the strategy adopted for transformer protection in power system. Discuss the challenges in the protection along with their merits and demerits. **(16)**

Q6 Describe the construction and principle of operation of SF_6 circuit breaker with neat sketches. Give a comparative analysis of SF_6 circuit breaker with that of air blast circuit breaker. **(16)**

Registration No.: | | | | | | | | | | |

Total Number of Pages : 02

Course: B.Tech
Sub_Code: REL6D001

6th Semester Regular/Back Examination: 2023-24
SUBJECT: Electric Power System Protection
BRANCH(S): ELECTRICAL
Time: 3 Hour
Max Marks: 100
Q.Code: P437

Answer Question No.1 (Part-1) which is compulsory, any eight from Part-II and any two from Part-III.
The figures in the right hand margin indicate marks.

Part-I

Q1 **Answer the following questions:** **(2 x 10)**

a) What do you mean by burden of relay and what is its unit?
b) What are the advantages of static relays over electromagnetic relays?
c) What do you mean by under reach and over reach of a relay?
d) An oil circuit breaker is rated at 1500 amps, 2000 MVA, 33kV, 3 sec, and 3 phases. Determine the making current and breaking current.
e) What is the function of a polarised relay?
f) What is the difference between time lag and instantaneous relays?
g) What do you understand by “Holding Ratio”?
h) What are the advantages and disadvantages of an Oil circuit breaker?
i) Explain what you mean by RRRV. Write its mathematical expression.
j) What are the applications of DC relays?

Part-II

Q2 **Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve)** **(6 × 8)**

a) Explain the construction and principle of operation of Shaded Pole Type non-directional relays. Comment on the merits and demerits of this type of relays.
b) Discuss the various components used in protective system and explain the basic principle of operation of electric power system protection.
c) Write a short note on the Current Chopping and Explain how to minimize its effect.
d) A 50 MVA, 11 kV, 3 phase alternator is subjected to different types of faults. The fault currents are as follows; 3-phase fault = 2000 Amps; Line-to-Line fault = 2600 Amps; Line-to-ground fault = 4200 Amps. The generator neutral is solidly grounded. Determine the values of the three sequence reactances of the alternator. Ignore the resistances.
e) What is meant by RRRV? Derive the expression for RRRV.
f) In short circuit test on a 3 pole, 132 kV circuit breaker, the following observations are made; p.f. of fault is 0.4, recovery voltage is 0.9 times the full line value, the breaking current is symmetrical, frequency of oscillations of restriking voltage is 16 kHz. Assuming neutral is grounded and fault is not grounded, determine the average RRRV.

g) Explain the principle of operation of directional over current relays with mathematical background. Explain what is meant by directional properties of the relays.

h) What is the need for Bus Bar Protection? How the Bus Bars protection scheme is realized. Comment on the stabilization of this type of protection scheme.

i) Explain the pilot protection scheme used in power system. Describe the different type of pilots with applications.

j) With a neat diagram explain the construction and principle of operation of Buchholz relays. Comment on the challenges on its design and operation.

k) Write a short note on SF6 Circuit Breakers.

l) Write a short note on the Coincidence type phase comparator.

Part-III

Only Long Answer Type Questions (Answer Any Two out of Four)

Q3 A 3-phase 20 MVA, 11kV star connected alternator has a synchronous reactance of 2.5 ohm/phase and resistance of 0.75 ohm/phase. It is being protected by a Merz-price balanced current system. Determine that what portion of the winding remains unprotected and if the neutral of the alternator is earthed through a resister of 5 ohms. Assume that the relay operates, when the out of balance current exceeds 25 % of the load current. **(16)**

Q4 Explain the construction and principle of operation of Balanced Beam type impedance relays and derive the mathematical equations for this type of relays. Discuss the opportunities and challenges faced for this type relays. **(16)**

Q5 Elaborate with neat diagram the strategy adopted for motor protection in power system. Discuss the challenges in the protection along with their merits and demerits. **(16)**

Q6 Describe the construction and principle of operation of Vacuum circuit breaker with neat sketches. Give a comparative analysis of Vacuum circuit breaker with that of air blast circuit breaker. **(16)**

Registration No.:

Total Number of Pages: 02

B.Tech/IDD (B.Tech and M.Tech)
REL6D001

6th Semester Regular/Back Examination: 2024-25
Electric Power System Protection
BRANCH(S): ELECTRICAL
Time: 3 Hour
Max Marks: 100
Q.Code : S205
Answer Question No.1 (Part-1) which is compulsory, any eight from Part-II and any two from Part-III.
The figures in the right hand margin indicate marks.

Part-I

Q1 Answer the following questions: **(2 x 10)**

a) What is RRRV? Explain its significance.
b) Identify the type of fault if the positive sequence component of voltage at the point of fault is zero.
c) What is meant by auto-reclosing action in power system protection?
d) What is the significance of sensitivity in the protection systems?
e) What is the impact of fault clearing time on the operation of protective devices?
f) What is the effect of CT saturation on the design of protective relaying systems?
g) What do you mean by under reach and over reach of a relay?
h) Draw the zero-sequence network for a Δ - Y connected 3-phase transformer.
i) What is meant by current chopping?
j) What do you mean by TSM? How it affects the performance of a relay?

Part-II

Q2 Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve) **(6 x 8)**

a) With a neat sketch, explain the recovery rate theory and energy balance theory of arc interruption in circuit breaker.
b) What is an impedance relay? Explain the operating principle, torque equation, and operating characteristics for impedance relay.
c) Write a short note on “Buchholz relay” used for the protection of transformers to detect incipient fault.
d) Discuss the working principle of coincidence type phase comparator? Derive the mathematical expressions related to it.
e) With a neat sketch explain the working of differential protection of 3-Phase circuits and balanced (opposed) voltage differential protection
f) What are the protective devices employed for the protection of an alternator against I) Overvoltage and II) Overspeed? Discuss them in brief.
g) Derive the general equations of magnitude comparator incorporating the block diagram and phasor diagram.

h) The current rating of an Overcurrent relay is 5 A. The relay has a plug setting of 150% and the time setting (TMS) of 0.4. The CT ratio is 400/5. Determine the operating time of the relay for a fault current of 6000 A. At TMS = 1, operating time at various PSM are given in the below table.

PSM	2	4	5	8	10	20
Operating Time (Sec)	10	5	4	3	2.8	2.4

i) With a neat sketch and waveform explain the interruption of capacitive current.

j) With a neat sketch, explain the construction and working principle of a directional or reverse power relay.

k) Define the term 'pilot' with reference to power line protection. List the different types of wire pilot protection schemes and explain for any one of the schemes.

l) Draw the schematic diagram of a numerical relay and explain the functions of various components.

Part-III

Only Long Answer Type Questions (Answer Any Two out of Four)

Q3 Explain with necessary diagram the principle of operation of a distance protection scheme. How a 3 zone stepped units help to achieve fast discrimination? **(16)**

Q4 Discuss the construction and general principle of operation of a transformer differential protection scheme? Explain percentage biasing of differential relay. **(16)**

Q5 Explain the construction and working principle of SF6 circuit breaker. Discuss the advantages and disadvantages of SF6 circuit breaker. **(16)**

Q6 Describe the construction and working principle of an Induction Cup Type Electromagnetic Relay. Derive the torque equation and discuss its advantages and disadvantages. **(16)**